ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Diversity in the Living World** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** a grass	Grasses show these features.
2	**(b)** helps judge distance for hunting	Forward eyes give depth perception.
3	**(c)** many plants will fail to pollinate	Pollination would fall.
4	**(c)** frog	Frogs show this.
5	**(a)** wide field of view to spot predators	Wide view.
6	**(b)** it provides more food, water and shelter	Rich resources support life.
7	**(c)** to avoid sinking in sand	Large area reduces pressure.
8	**(b)** being active at night	Avoid heat.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R is a general pattern and does not explain why wheat has fibrous roots.
10	**(a)** Both A and R are true and R is the correct explanation of A.	The smooth, tapering body reduces resistance in water, so R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Green stem carries out photosynthesis (1); Spines reduce water loss (1).	1+1
12	It breathes air through lungs (1); It gives birth to young and feeds milk (1).	1+1
13	Water supports them (1); Flexible stems bend with the flow of water without breaking (1).	1+1
14	No (1); Bats and insects also fly (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tree (1); Tap root goes deep (1); Net veins in leaves (1).	1+1+1
16	Colour can vary within a group (1); Unrelated animals may have the same colour (1); Features like body covering and habitat are better (1).	1+1+1
17	Beak shape matches food (1); Woodpecker has strong beak for wood (1); Hummingbird has thin beak for nectar (1).	1+1+1
	OR: Duck: webbed feet (1); Fish: fins and tail (1); Fish: streamlined; duck: oily feathers (1).	1+1+1
18	Stores fat (1); Fat gives energy when food is scarce (1); Body fat in one place keeps the rest cooler (1).	1+1+1
	OR: Flowering is a feature of many types (1); Herbs, shrubs, trees all flower (1); Size and stem differ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Mango (1). (ii) Pumpkin (1). (iii) Herb: small with soft, green stem; shrub: bushy with a hard stem branching near the base (1). (iv) A creeper spreads on the ground with a weak stem (1); a climber climbs up a support with tendrils or other parts; example: grapevine (1).	1+1+1+2
20	Hollow bones: lightness (1); Feathers: flight and warmth (1); Beak: food type (1); Feet: perching, swimming or hunting (1); Examples: duck, eagle, parrot (1).	1+1+1+1+1
	OR: Stem: soft, woody, thick (1); Height: low, medium, tall (1); Branching differs (1); Lifespan generally longer in trees (1); Uses such as vegetables, hedges, timber (1).	1+1+1+1+1

Mistakes to avoid:

- Calling any small plant a herb; a shrub is also small but has a **hard stem** branching near the base.
- Mixing up creepers and climbers: creepers spread on the ground, climbers go up a support.
- Writing that tap roots go with parallel venation; tap root usually goes with **reticulate** venation, fibrous with parallel.
- Confusing **habitat** (the place) with **adaptation** (the feature).
- Saying a camel stores water in its hump; the hump stores fat.
- Thinking venation means the roots; it is the pattern of veins in the **leaf**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Diversity in the Living World** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** prop roots	They support the tree.
2	**(d)** to store water	Stem stores water.
3	**(c)** its gills cannot take oxygen from air	Gills work in water.
4	**(c)** to repel water	Water rolls off.
5	**(b)** birds	Feathers define birds.
6	**(d)** bat	Milk feeding defines mammals.
7	**(b)** to show similarities useful for study	Classification organises.
8	**(c)** whether it has a thick woody trunk and is tall	Trunk and height.
9	**(c)** A is true but R is false.	A is true; R is false because cactus leaves are small spines.
10	**(d)** A is false but R is true.	A is false: that describes a tree. R correctly describes a shrub.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It spreads along the ground quickly (1); It needs less energy for making a trunk (1).	1+1
12	Moist skin needs water (1); They lay eggs in water (1).	1+1
13	Fat stored in the hump is used (1); They lose very little water (1).	1+1
14	White colour helps in hunting unseen (1); Fur and fat keep it warm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Snow slides off (1); Needles lose less water (1); Withstand strong cold winds (1).	1+1+1
16	Fibrous roots bind the soil (1); Leaves cover the ground (1); Prevent erosion by wind and water (1).	1+1+1
17	Plants are overeaten (1); Food shortage follows (1); Carnivores may increase (1).	1+1+1
	OR: Thin leaves absorb gases from water (1); Roots absorb water directly so are small (1); Flexible stems bend with water flow (1).	1+1+1
18	Loss of homes (1); Shortage of food (1); Increased danger from humans (1).	1+1+1
	OR: Its body features help it survive heat and shortage of water (1); Hump stores fat (1); Wide feet do not sink in sand (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Fibrous roots (1). (ii) Mustard (1). (iii) Venation (1). (iv) Reticulate venation (1); carrot (or mango, gram) (1).	1+1+1+2
20	Food chains (1); Medicines from plants (1); Forests influence rainfall (1); Tourism (1); Loss leads to imbalance (1).	1+1+1+1+1
	OR: Camouflage hides the animal in its surroundings (1); Chameleon blends with leaves (1); Stick insect looks like a twig (1); Predators fail to notice the animal (1); So it survives (1).	1+1+1+1+1

Mistakes to avoid:

- Calling any small plant a herb; a shrub is also small but has a **hard stem** branching near the base.
- Mixing up creepers and climbers: creepers spread on the ground, climbers go up a support.
- Writing that tap roots go with parallel venation; tap root usually goes with **reticulate** venation, fibrous with parallel.
- Confusing **habitat** (the place) with **adaptation** (the feature).
- Saying a camel stores water in its hump; the hump stores fat.
- Thinking venation means the roots; it is the pattern of veins in the **leaf**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Diversity in the Living World** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** long thin beak	Probing beak.
2	**(a)** cactus	Cactus has water-saving features.
3	**(d)** webbed feet	Webbed feet are for swimming.
4	**(c)** India has a rich variety of plants and animals	India is among the richest regions in biodiversity.
5	**(c)** a thick trunk and deep roots	Strong anchorage.
6	**(a)** woody stem branching near the base	Woody and branched.
7	**(d)** they have soft stems and are usually short	Herbs are short.
8	**(a)** climber	Tendrils help climbing.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Being tender and non-woody is what makes the stem soft and green, so R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but the type of venation does not decide whether a plant is a tree.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Shrub branches near the base (1); Tree has a single main trunk (1).	1+1
12	Hawk has hooked beak to tear flesh (1); Duck has flat beak to filter food in water (1).	1+1
13	More rain and warmth allow many kinds of plants to grow (1); Different plants find suitable conditions in different parts of the forest (1).	1+1
14	Many species live there (1); Destroying habitats leads to extinction (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Observe the veins (1); Parallel veins suggest a grass-like plant (1); Check the root type if available (1).	1+1+1
16	To collect more light (1); To see in dark (1); Helps in hunting at night (1).	1+1+1
17	Gills take oxygen dissolved in water (1); Lungs take oxygen from air (1); Frog also uses its skin (1).	1+1+1
	OR: Fish lose food and shelter (1); Oxygen supply reduces (1); Food chain breaks (1).	1+1+1
18	Spinach for leaves (1); Wheat for grain (1); Tomato for fruit (1).	1+1+1
	OR: Roots seek water (1); Stems seek light (1); This helps survival (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Habitat (1). (ii) Broad padded feet (1). (iii) Gills (1). (iv) Adaptation is a feature that helps an organism live in its habitat (1); a camel has no gills or fins, so it cannot breathe or swim in water (1).	1+1+1+2
20	Arctic: very cold (1); Polar bear: fur and fat (1); Desert: hot, dry (1); Camel: hump, wide feet (1); Different adaptations (1).	1+1+1+1+1
	OR: Walk and note plants (1); Classify into herbs, shrubs, trees (1); Note leaves and roots (1); Record in a table (1); Draw a conclusion about diversity (1).	1+1+1+1+1

Mistakes to avoid:

- Calling any small plant a herb; a shrub is also small but has a **hard stem** branching near the base.
- Mixing up creepers and climbers: creepers spread on the ground, climbers go up a support.
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